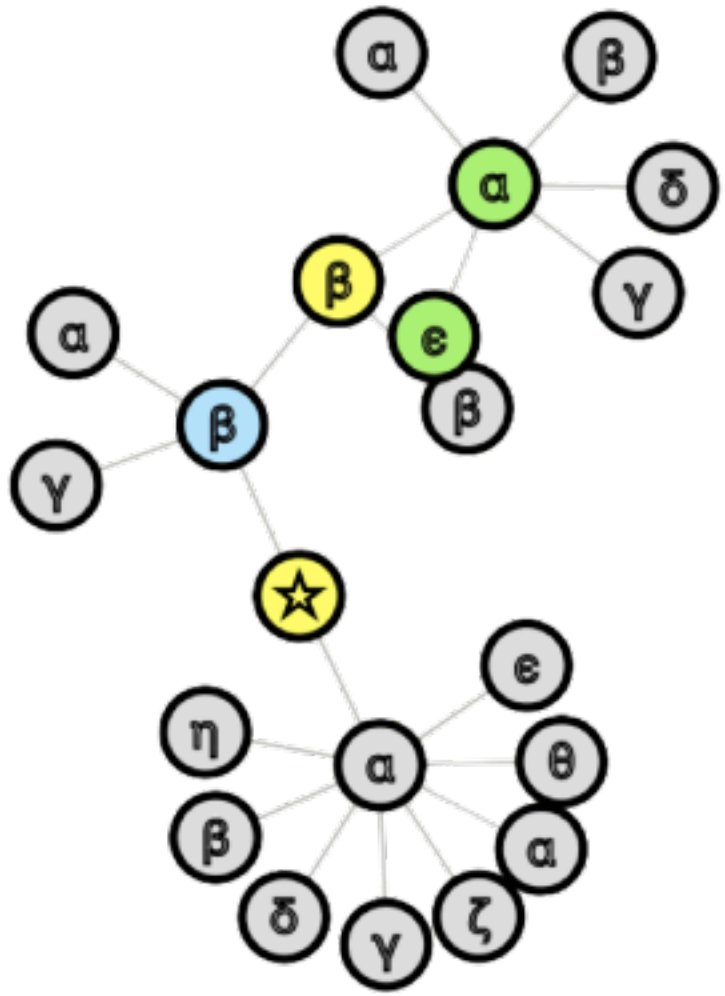
Recursive Delegation

A framework for recursive multi-agent systems

- Recursive Delegation: start with a single agent, which has a tool to delegate tasks to new child agents

- When presented with a complex task, the agent can decompose it and give each part to a new child agent

- Each child can complete their own sub-task if it's simple, or decompose further and delegate again



Delegation Graph

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EMNLP 2024 Demo

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Recursive Delegation with Kani

Using other LLMs as tools

```
class RecursiveMultiAgentKani(Kani):
  @ai_function()
  async def delegate(self, instructions: str):
    """Delegate a smaller part of the current task to a helper agent."""
    sub_agent = RecursiveMultiAgentKani(engine)
    result = None
    async for msg in sub_agent.full_round(instructions):
      # save the sub-agent's final result to return to the parent
      result = msg.text
    return result

  # additional tools for each agent...

root_agent = RecursiveMultiAgentKani(engine)
chat_in_terminal(root_agent)
```